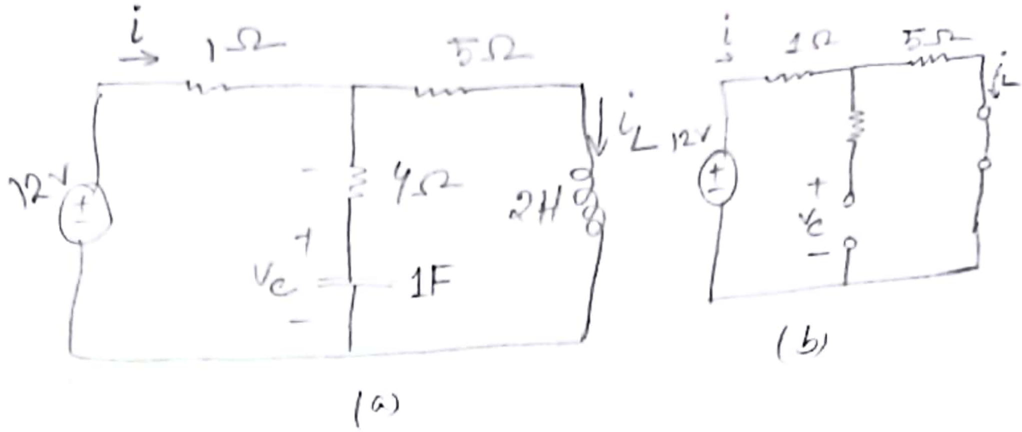
Example 6.10 :



Consider the circuit in Fig 6.27(a). Under dc conditions, find : (a) i, v_c and i_L (b) the energy stored in the capacitor and inductor.

Solution:

(a)

Here, $i = i_L$

$R = (1+5)\,\Omega = 6\,\Omega$

$i = \frac{V}{R} = \frac{12v}{6\Omega} = 2A$

$\therefore i_L = 2A$

The voltage v_c is the same as the voltage across the 5Ω resistor. Hence,

$v_c = 5i = 5 \cdot 2 = 10V$

(b) The energy in the capacitor is,

$$W_c = \frac{1}{2} C V_c^2$$

$$= \frac{1}{2}(1)(10)^2$$

$$= 50j$$

The energy in the inductor is,

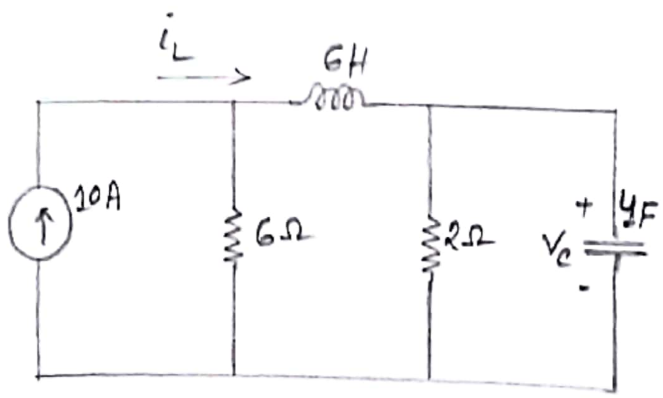
$$W_L = \frac{1}{2} L i_L^2$$

$$= \frac{1}{2}(2)(2)^2$$

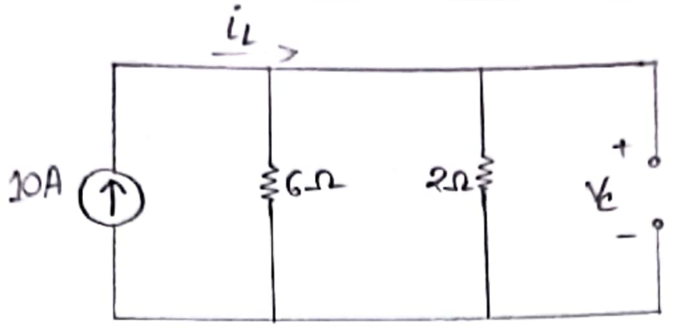
$$= 4j$$

Practice Problem-6.10:

Determine V_c, i_L and the energy stored in the capacitor and inductor in the circuit under dc conditions.



Solution:



$$i_L = 10 \times \frac{6}{6+2} = \frac{60}{8} = 7.5A$$

$$V_c = 7.5 \times 2 = 15\,V$$

Energy of Capacitor:

$$W_c = \frac{1}{2} C V_c^2 = \frac{1}{2}(4)(15)^2 = 450j$$

Energy of Inductor:

$$W_L = \frac{1}{2} L i_L^2 = \frac{1}{2}(6)(7.5)^2 = 168.75j$$

Example - 9.1:

Find the amplitude, phase, period and frequency of the sinusoid.

$$v(t) = 12\cos(50t + 10^\circ)$$

Solution:

Compare $v(t) = 12\cos(50t + 10^\circ)$ with $v(t) = V_m \cos(\omega t + \phi)$

We get,

Amplitude, $V_m = 12\,V$

phase angle, $\phi = 10^\circ$

Angular frequency, $\omega = 50$ rad/s

Time period, $T = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{50} = 0.1257\,s$

Frequency, $f = \dfrac{1}{T} = \dfrac{1}{0.1257} = 7.958\,Hz$

Practice Problem-9.1:

Given the sinusoid $30\sin(4\pi t - 75^\circ)$, calculate its amplitude, phase, angular frequency, period and frequency.

Solution:

$$V(t) = 30\sin(4\pi t - 75^\circ)$$

Amplitude, $V_m = 30\,V$

Phase angle, $\phi = -75^\circ$

Angular frequency, $\omega = 4\pi = 12.57\ rad/s$

Time period, $T = \frac{2\pi}{\omega} = 0.5\,s$

frequency, $f = \frac{1}{T} = 2\,Hz$

Example - 9.2:

Calculate the phase angle between $v_1 = -10\cos(\omega t + 50^\circ)$ and $v_2 = 12\sin(\omega t - 10^\circ)$. State which sinusoid is leading.

Solution:

$$V_1 = -10\cos(\omega t + 50^\circ)$$

$$V_2 = 12\sin(\omega t - 10^\circ)$$

Convert V_1 into sin form,

$$V_1 = 10\sin(\omega t + 50^\circ - 90^\circ)$$

$$= 10\sin(\omega t - 40^\circ)$$

$$V_2 = 12\sin(\omega t - 10^\circ)$$

From V_1 and V_2,

$$\cancel{10^\circ} \quad -10^\circ > -40^\circ$$

Hence the angle of V_2 is greater than the angle of V_1.

$\therefore$ V_2 ~~is~~ leads V_1 by 30°.

Practice Problem - 9.2:

Find the phase angle between

$$i_1 = -4\sin(377t + 55^\circ) \text{ and } i_2 = 5\cos(377t - 65^\circ)$$

Does i_1 lead or lag i_2?

Solution:

Express i_1 in cosine form,

$$i_1 = 4\cos(377t + 55^\circ + 90^\circ)$$

$$= 4\cos(377t + 145^\circ)$$

$$i_2 = 5\cos(377t - 65^\circ)$$

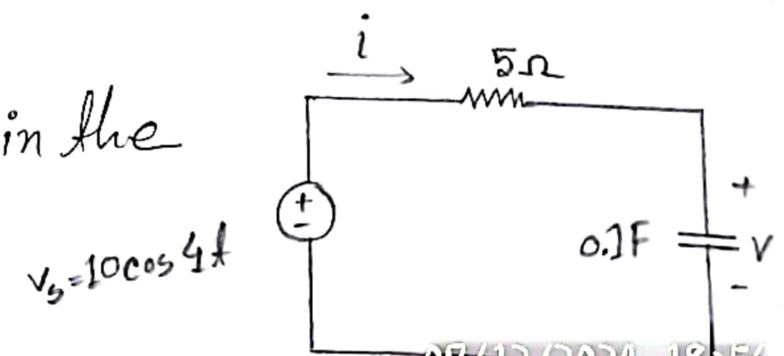
From i_1 & i_2, $145^\circ > -65^\circ$

Hence $i_1 > i_2$. $\therefore$ Phase angle $= 145^\circ - (-65^\circ)$
$= 210^\circ$

$\therefore$ i_1 leads i_2 by $\cancel{145^\circ}$ 210°.

Example - 9.9:

Find $V(t)$ and $i(t)$ in the circuit.



Solution:

$V_s = 10\angle 0^\circ$

$\omega = 4$

Impedence, $Z = 5 + \frac{1}{j\omega C}$

$= 5 + \frac{1}{j\,4 \times 0.1}$

$= 5 - j2.5\ \Omega$

$I = \frac{V_s}{Z} = \frac{10\angle 0^\circ}{5 - j2.5} = \frac{10\angle 0^\circ}{5.59\angle -26.56^\circ}$

$= 1.789\angle 26.56^\circ\ A$

$V = I Z_c = I \times \frac{1}{j\omega C} = \frac{1.789\angle 26.56^\circ}{j4 \times 0.1}$

$= \frac{1.789\angle 26.56^\circ}{0.4\angle 90^\circ}$

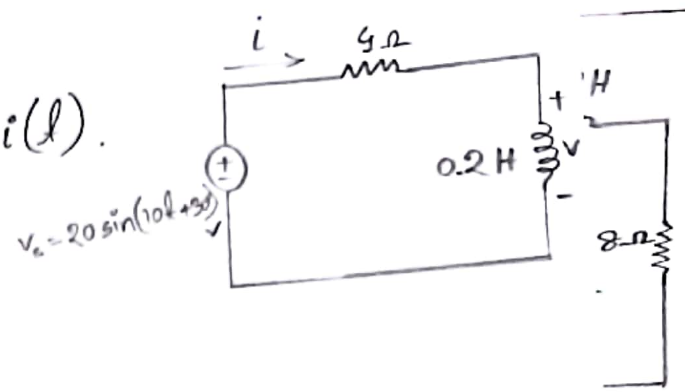
$= 4.46\angle -63.43^\circ\ V$

$\therefore\ V(t) = 4.46\cos(4t - 63.43^\circ)\,V$

$i(t) = 1.789\cos(4t + 26.56^\circ)\,A$

Practice Problem: 9.9:

Determine $v(t)$ and $i(t)$.



Solution:

$$V_s = 20\angle 30^\circ$$

$$\omega = 10$$

$$Z = 4 + j\omega L = 4 + (j10 \times 0.2)$$

$$= 4 + 2j$$

$$= 4.47\angle 26.56^\circ$$

$$I = \frac{V_s}{Z} = \frac{20\angle 30^\circ}{4.47\angle 26.56^\circ} = 4.47\angle 3.44^\circ$$

$$V = I Z_L = (4.47\angle 3.44^\circ) \times j\omega L$$

$$= (4.47\angle 3.44^\circ) \times j10 \times 0.2$$

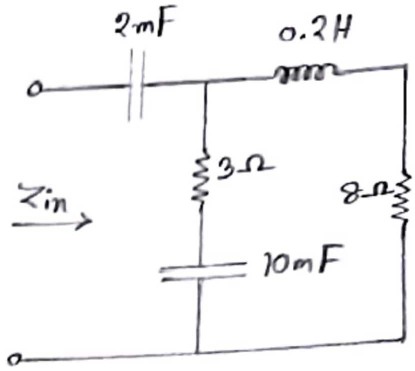
$$= 8.94\angle 93.44^\circ$$

$$\therefore\ v(t) = 8.94\sin(10t + 93.44^\circ)\ V$$

$$I(t) = 4.47\sin(10t + 3.44^\circ)\ A$$

Example - 9.10:

Find the input impedance of the circuit. Assume that the circuit operates at $\omega = 50 \text{ rad/s}$.



Solution: $\omega = 50 \text{ rad/s}$

$$Z_{C_1} = \frac{1}{j\omega C_1} = \frac{1}{j50 \times 2 \times 10^{-3}} = \cancel{-10j}\ j(-10)$$

$$Z_{C_2} = \frac{1}{j\omega C_2} = \frac{1}{j50 \times 10 \times 10^{-3}} = \cancel{-2j}\ -j2$$

$$Z_L = j\omega L = j50 \times 0.2 = \cancel{10j}\ j10$$

Now,

$$Z_1 = Z_L + 8\Omega = \cancel{10j+}\ (8 + 10j)\Omega$$

$$Z_2 = 3\Omega + Z_{C_2} = (3 + -j2)\Omega$$

$$Z_3 = Z_1 \| Z_2 = \frac{(8+10j)(3-j2)}{8+j10+3-j2}$$

$$= \frac{(8+10j)(3-j2)}{11+8j}$$

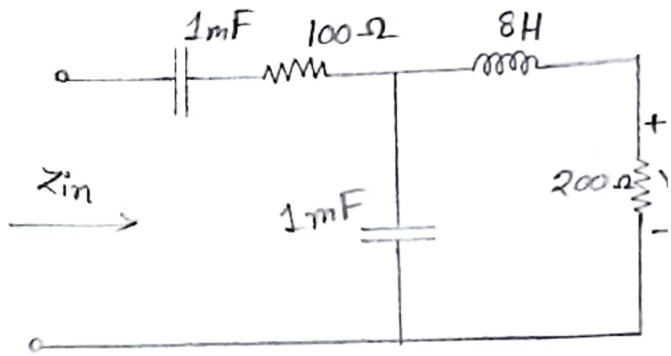
$$= \frac{44+14j}{11+8j}$$

$$Z_{in} = Z_{C_1} + Z_3 = -j10 + \frac{44+14j}{11+8j}$$

$$= (3.22 - j11.07)\ \Omega$$

Practice Problem - 9.10:

Determine the input impedance of the circuit at $\omega = 10$ rad/s.



Solution: $\omega = 10$ rad/s

$$\cancel{Z_{C_{10mF}} = \frac{1}{j\omega C} = \frac{1}{j10 \times 10\times10^{-3}} =}$$

$$Z_c = \frac{1}{j\omega C} = \frac{1}{j10 \times 1\times10^{-3}} = -100j$$

$$Z_1 = 100 + \frac{1}{j10 \times 1 \times 10^{-3}} = 100 - 100j$$

$$Z_2 = 200 + (j10 \times 8) = 200 + 80j$$